

Flashcard Study App

◆ Overview

- Vanilla JavaScript based flashcard application
 - Supports interactive learning with animations and tracking
 - Data stored using browser localStorage
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◆ Core Features

-  Create flashcards dynamically (question & answer)
 -  Flip animation to reveal answers
 -  Navigate using Previous / Next buttons
 -  Add categories for organization
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◆ Advanced Features

-  Shuffle cards for random study
 -  Score tracking (correct / incorrect)
 -  Mark cards as Known / Unknown
 -  Category filtering
 -  Data saved using localStorage
 -  Responsive design (mobile + desktop)
 -  Keyboard shortcuts for fast use
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◆ How to Use

Getting Started

- Open index.html in browser
- Enter question & answer
- Add category (optional)
- Click **Add Card**

Studying

- Click card / press **Space** → Flip
- Press → / ← → Navigate
- Press:
 - **Enter** → Known

- **Backspace** → Unknown

◆ Additional Features

- Shuffle cards
- Filter by category
- Reset score
- Clear all flashcards

◆ Keyboard Shortcuts

Key	Action
Space	Flip card
→	Next card
←	Previous card
Enter	Mark Known
Backspace	Mark Unknown

◆ Technologies Used

- HTML5 → Structure
- CSS3 → Animations & layout
- JavaScript → Logic (no frameworks)

◆ Data Structure (Flashcard)

```
{  
  id: 1234567890,  
  question: "Capital of France?",  
  answer: "Paris",  
  category: "Geography",  
  isKnown: false,  
  createdAt: "2024-01-15"  
}
```

◆ Key Implementations

- 3D flip animation using CSS transforms

- `LocalStorage` for saving data
 - Centralized state using JavaScript class
 - Filtering system for categories
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◆ File Structure

Flashcard Study App/

```
|— index.html
|— styles.css
|— script.js
|— README.md
```

◆ Browser Support

- Chrome 
 - Firefox 
 - Safari 
 - Edge 
 - IE 
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◆ Responsive Design

- Desktop → Two-column layout
 - Tablet → Single column
 - Mobile → Optimized layout
 - Small mobile → Compact view
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◆ Tips for Learning

- Start with small number of cards
 - Use categories
 - Review regularly
 - Shuffle frequently
 - Practice unknown cards more
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◆ Data Storage

- Stored in **`localStorage`**

- Key: flashcardAppData
 - Not shared across browsers/devices
 - Not saved in incognito mode
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◆ **Troubleshooting**

- Data not saving → Enable localStorage
 - Animation lag → Update browser
 - Mobile UI shift → Normal behavior
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◆ **Future Enhancements**

- Image support
 - MCQ options
 - Spaced repetition system
 - Study statistics
 - Import/export (JSON)
 - Dark mode
 - Multi-language
 - Cloud sync
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◆ **License**

- Free for personal & educational use
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♥ **Conclusion**

- Simple, interactive, and efficient learning tool
- Helps improve memory using active recall